College of Engineering, SQU

Sultan Qaboos University
College of Engineering
Office of Assistant Dean for
Industrial Training and Community Services

Weekly Report
Week #: 3
Date: 14/07/2026

Student	Full Name: Malek Al-shukily	ID: 141776
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Supervisor	Full Name: MODES	Email:
	Training Company & Department: Mohammed Muhmoed Al-Sahhi	Tel./Mobile: 95050090

Tasks Completed (Briefly Describe any tasks that you completed during this week):

- ~~otser~~ studied the design, safety systems, and operating
- Principles of fuel stations.
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Tasks in Progress (if any):

- Preparing pre-study notes for the upcoming
- saftey and fire department rotation
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Plan for next week (if different from this week):

- Rotate to the Fire, Health, safety, and Environment
- (HSE) department.
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Problems/Challenges/Recommendations (if any):

No major problem we phase this week.

Supervisor Comments (your feedback is very important for our continuing improvement):

the suden do all the besa during the training program
and he acheived all his responsipiltes

Student:
I have read and understood the whole content of the Students' Training Manual.

Signature: ....

Supervisor:
I will do my best to guide the student towards achieving all required deliverables.

Signature: ....

Ministry of Defence - Engineering Services Directorate
RAFO BULK FUEL INSTALLATION

Note: A copy of this report should be faxed to 24413416 or scanned and emailed to your SQU supervisor on a weekly basis. Originals should be placed in the Appendix of the **Final Report.**

Weekly Training Report

Industrial Training & Field Operations Experience

TRAINING WEEK	LOCATION / FACILITY	REPORT DATE
Week 3 (Current Week)	Bulk Fuel & LPG Facilities	July 14, 2026

1. EXECUTIVE SUMMARY

During the third week of the field training program, the primary focus shifted towards large-scale fuel infrastructure, specifically bulk fuel facilities, commercial petrol distribution stations, and Liquefied Petroleum Gas (LPG) storage systems. This week provided a practical, hands-on understanding of how fuel is safely stored, managed, and distributed to meet high market demands, while strictly adhering to safety and environmental standards.

2. KEY LEARNING AREAS & FIELD ACTIVITIES

A. Petrol Stations & Fuel Distribution Systems

We thoroughly investigated the technical anatomy of retail fuel stations, starting from the underground storage tanks (USTs) to the fuel dispensers. Key activities included:

- **Inventory & Tank Monitoring:** Understood the automatic tank gauging (ATG) systems used to track fuel levels, detect water contamination, and monitor fuel temperature in real-time.
- **Submersible & Dispenser Pumps:** Studied the hydraulics behind fuel delivery, comparing pressurized systems (using submersible pumps) with suction-based dispensing units.
- **Vapor Recovery Systems (Stage I & II):** Learned how toxic fuel vapors are captured during delivery from bulk tankers to underground tanks, as well as during vehicle refueling, significantly reducing emission leaks into the atmosphere.

B. Bulk LPG Storage & Safety Systems

The training also deep-dived into the engineering of massive LPG storage facilities, which handle pressurized liquefied gas. We focused on:

- **Storage Tank Design:** Examined both mounded and above-ground LPG bullet tanks, studying how they handle extreme pressure and thermal expansion.
- **Pressure Regulation & Valves:** Analyzed the safety relief valves (SRVs) and automatic shut-off systems designed to prevent catastrophic overpressurization or gas leaks.

- **Fire Protection & Deluge Systems:** Observed the automated water spray/deluge systems designed to cool LPG vessels rapidly in the event of an adjacent fire hazard.

> **Critical Reflection: Safety First (HSE)**
>
> Operating within high-hazard fuel zones highlighted the absolute necessity of strict Hazardous Area Classification (Zones 0, 1, and 2). We practiced using non-sparking tools, understood static electricity mitigation through proper earthing, and participated in emergency isolation shutdown protocols.

3. KEY TAKEAWAYS & COMPETENCIES GAINED

This week successfully bridged theoretical thermodynamic and fluid mechanics concepts with actual industrial applications. Key skills acquired include the ability to interpret Piping and Instrumentation Diagrams (P&IDs) for fuel flow, run basic diagnostics on automatic level gauges, and evaluate the safety readiness of pressurized gas installations.

Trainee Signature	**Site Supervisor Approval**
Field Engineering Intern	Lead Facilities Engineer